HDOC DAY – 61

Algorithm

1. Start.
2. Read the size of the array.
3. Declare a variable-length integer array nums[size].
4. Read the array elements.
5. Read target.
6. Pass the array address, size, and target to two_sum().
7. Using the outer loop, select each element from the array.
8. Using the inner loop, compare it with every element after it.
9. Check whether the sum of the two elements is equal to target.
10. If equal, display their indices and return from the function.
11. If no pair is found after completing both loops, display -1 -1.
12. Stop.

C PROGRAM:

```
#include <stdio.h>


void two_sum(int *nums, int size, int target)
{
   int i, j;


   for (i = 0; i < size - 1; i++)
   {
      for (j = i + 1; j < size; j++)
      {
```

```
            if (*(nums + i) + *(nums + j) == target)
            {
                printf("%d %d\n", i, j);
                return;
            }
        }
    }

    printf("-1 -1\n");
}

int main()
{
    int size, i, target;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter the array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", &nums[i]);

    printf("Enter the target: ");
```

```
    scanf("%d", &target);

    two_sum(nums, size, target);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Array	Target	Expected Output
1	2 7 11 15	9	0 1
2	3 2 4	6	1 2
3	3 3	6	0 1
4	1 2 3	7	-1 -1
5	1 5 8 10	15	1 3